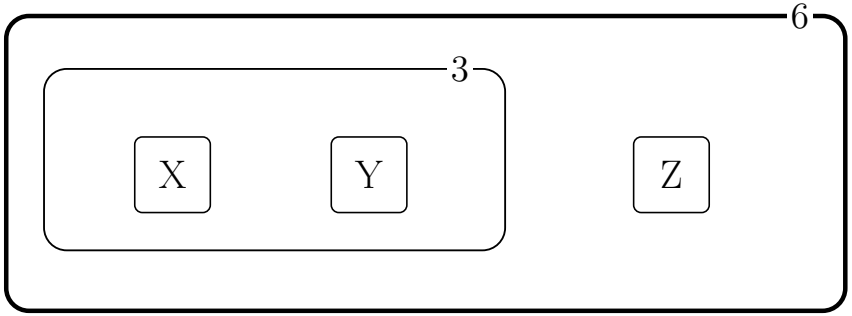
Groups can sit inside one another or be separate. They never cross. One group contains all the cards. Each card is also a one-card group. A group may contain three or more smaller groups. Every larger group has a positive label, and labels increase as you go outward. A group's size and number of cards do not determine its label.

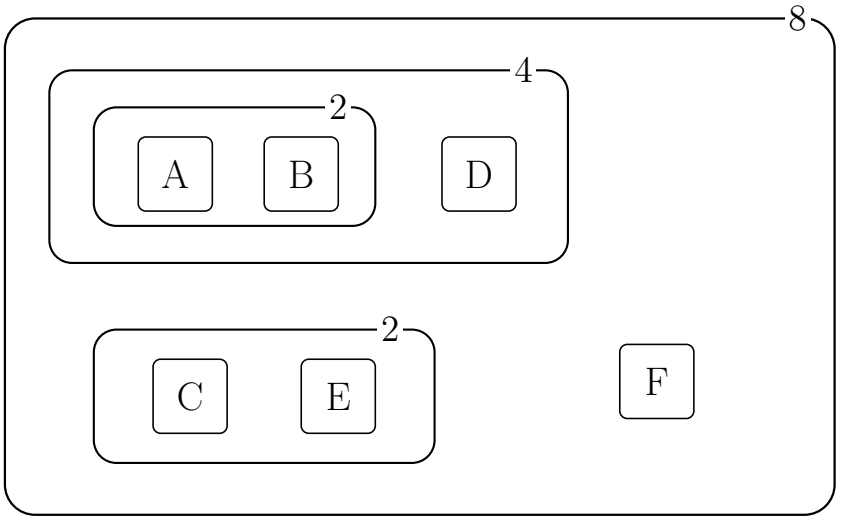
The distance between two different cards is the label of their smallest common group. A card's distance from itself is 0. We write A–B for the distance between A and B.

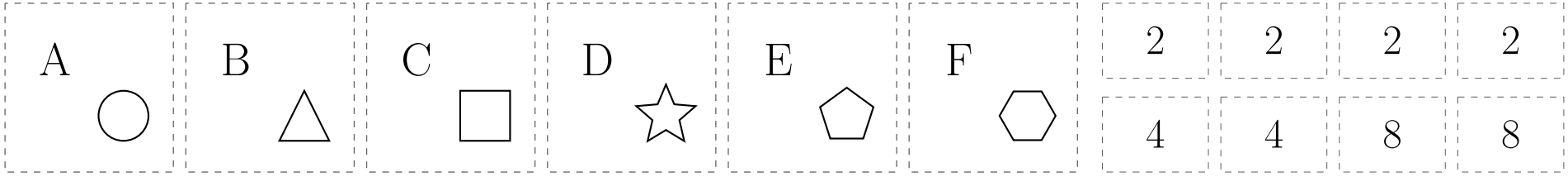


X and Z ⟶ the thick group is their smallest common group ⟶ X–Z = 6.

X–Y = 3. X–X = 0.

Problem 1: Build this world with cards A through F and loops. Choose six pairs and record their distances. Rearrange the cards and loops so every distance stays the same.





Problem 2: Use A, B, C and labels 2, 4, 8. Make each set of distances with loops, or cross it out if it cannot be made. Choose one crossed-out set and explain why no arrangement works.

A–B = 2 A–C = 2

B–C = 2

A–B = 2 A–C = 4

B–C = 4

A–B = 2 A–C = 2

B–C = 4

A–B = 4 A–C = 4

B–C = 4

A–B = 2 A–C = 4

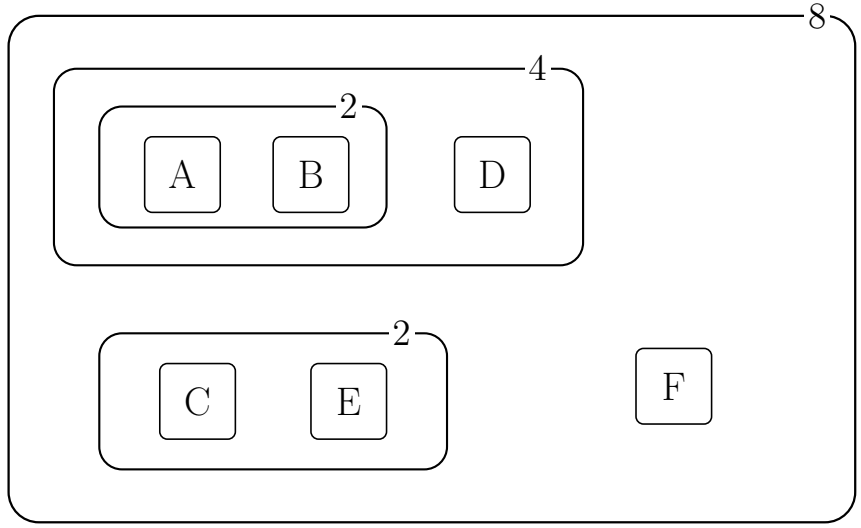
B–C = 8

A–B = 2 A–C = 8

B–C = 8

Problem 3: A ball of radius r around a card contains that card and every card at distance at most r. In the example on page 1, radius 1 around X gives just X; radius 4 around X gives X and Y.

In this world, find the radius-3 balls around A, C, D and F. Then find all different balls that contain B, using radii 0, 1, 2, 3, 4, 7 and 8. Draw each different collection once and write the radii that give it.



Problem 4: Use cards A through E to make a world that fits all ten clue cards. Use labels 2, 4, 8. Your partner checks each clue against the loops. Draw two different-looking worlds with exactly these distances.

A–B = 2	A–C = 4	B–C = 4	D–E = 2	A–D = 8
A–E = 8	B–D = 8	B–E = 8	C–D = 8	C–E = 8